Neurocrit Care (2021) 34:301–311
https://doi.org/10.1007/s12028-020-00989-1

NEUROCRITICAL CARE SOCIETY

REVIEW ARTICLE

Drug Dosing in Patients Undergoing Therapeutic Plasma Exchange

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Abstract

Therapeutic plasma exchange (TPE) is an extracorporeal process in which a large volume of whole blood is taken from the patient's vein. Plasma is then separated from the other cellular components of the blood and discarded while the remaining blood components may then be returned to the patient. Replacement fluids such as albumin or fresh-frozen plasma may or may not be used. TPE has been used clinically for the removal of pathologic targets in the plasma in a variety of conditions, such as pathogenic antibodies in autoimmune disorders. TPE is becoming more common in the neurointensive care space as autoimmunity has been shown to play an etiological role in many acute neurological disorders. It is important to note that not only does TPE removes pathologic elements from the plasma, but may also remove drugs, which may be an intended or unintended consequence. The objective of the current review is to provide an up-to-date summary of the available evidence pertaining to drug removal via TPE and provide relevant clinical suggestions where applicable. This review also aims to provide an easy-to-follow clinical tool in order to determine the possibility of a drug removal via TPE given the procedure-specific and pharmacokinetic drug properties.

Keywords: Plasmapheresis, TPE plasma exchange, Drug removal, Apheresis

Introduction

Therapeutic plasma exchange (TPE), also referred to as plasmapheresis, is an extracorporeal process in which a large volume of whole blood, taken from the patient's vein, is separated into its various blood components through filtration or centrifugation. The plasma is then separated from the other cellular components of the blood and discarded while the remaining blood components may then be returned to the patient with or without replacement fluids such as albumin or fresh-frozen plasma (FFP) [1]. The filtration method uses a microporous filter, which allows plasma to pass through, filtering it from other cellular components found in whole blood. Conversely, centrifugation uses a device which separates the whole blood into layers based on density: plasma, platelets, lymphocytes, granulocytes and erythrocytes [1]. The centrifugation device allows for selective extraction of any of these layers, leading to other types of therapeutic apheresis in which blood components other than plasma are the target for removal [2].

TPE has been used clinically since the early 1970s for the removal of pathologic targets in the plasma in a variety of conditions, such as pathogenic antibodies in autoimmune disorders [3]. TPE is becoming more common in the neurointensive care space as autoimmunity has been shown to play an etiological role in many acute neurological disorders. It is important to note that not only does TPE removes pathologic elements from the plasma, but may also remove drugs and coagulation factors, which may be an intended or unintended consequence [4]. Inadequate drug concentrations can be a serious unintended consequence for patients treated with medications in which sustained adequate drug concentrations are vital, such as with many antimicrobials. The removal of coagulation factors by TPE may lead to bleeding or thrombosis and may further cause clinical downstream

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effects in medications that rely on these factors. However, the degree of drug and coagulation factor removal and its implication on potential dosing adjustments is still not yet well understood, as not all drugs are equivalent with regard to their removal by TPE. There are many factors that can affect drug removal in patients undergoing TPE; however, the scarcity of quality evidence and conflicting conclusions stated in case reports and case series can lead to confusion and uncertainty among clinicians. There have been few reviews of the existing literature of the effects of TPE on drug disposition; however, there are no definitive guidelines or manufacturers' recommendations regarding drug removal in patients undergoing TPE [5–8].

The objective of the current review is to provide an up-to-date summary of the available evidence pertaining to drug removal via TPE and provide relevant clinical suggestions where applicable. This review also aims to provide an easy-to-follow clinical tool in order to determine the possibility of a drug removal via TPE given the procedure-specific and pharmacokinetic drug properties.

Methods of Literature Search

A literature search of the databases MEDLINE (1946–April 2019), PubMed (1973–February 2019) and EMBASE (1974–April 2019) was completed using the following search terms: "Pharmacokinetics," "Drug Clearance," "Drug Dialysability," "Drug Distribution," "Drug Elimination," "Drug Excretion," "Plasma Concentration–Time Curve," "Drug Overdose" and "Drug Monitoring." These terms were combined with the following search terms: "Plasmapheresis" and "Plasma Exchange." Title and Abstract screening were performed to eliminate irrelevant articles and duplicates. A full text review was performed to further refine the search results. Books, reviews, animal studies and editorials were excluded. In addition, a manual search for additional relevant studies was performed by reviewing the reference lists of the relevant reviews and the selected studies. Data collected included drug name, study type (overdose versus therapeutic), study design, time from last dose to TPE, drug concentration change and fraction eliminated from the body as a result of TPE and TPE duration.

Summary of Evidence

A total of 125 studies were included in this review. The information presented was categorized by therapeutic drug classes and summarized in Tables 1, 2, 3, 4, 5 and 6 along with the percent protein binding (f_b %), volume of distribution (V_d) and half-life for each drug as a reference. Approximately, one-third of the reports were about drug removal by TPE during therapy, while approximately two-thirds were about using TPE during drug overdose. It is important to consider that in overdose situations, drug pharmacokinetics may be significantly different than in normal dosing conditions [9] and much of the literature reported on this subject is related to overdose situations. Therefore, caution is recommended when applying previous findings to current clinical scenarios if the patient in question has a significantly different clinical status than those previously reported.

Factors Affecting Drug Removal by TPE

The degree of drug removal by TPE is affected by both the drug pharmacokinetic characteristics and TPE-specific properties. These factors and their relevance to the extent of drug removal are summarized in the following sections.

Pharmacokinetics

Distribution: Protein Binding, Volume of Distribution and Multi-compartmental Kinetics

The most literature regarding drug removal via TPE highlights the importance of two key pharmacokinetic factors affecting drug removal: volume of distribution and plasma protein binding [5]. Additionally, the kinetic model that the drug in question displays can result in differing consequences regarding the amount of drug removed during TPE.

Volume of distribution (V_d) is the apparent volume in which the drug distributes in the body. In other terms, the higher the V_d is, the larger the amount of the drug that is distributed to tissues. Drugs that exhibit a low V_d (< 0.2–0.3 L/kg), correlating to a higher drug plasma concentrations, are more susceptible to removal by TPE [10]. Changes in body volumes, due to factors such as critical illness or hydration status, affect drug distribution and protein binding and, hence, propensity to TPE removal. Additionally, drugs that are distributed into tissues may display post-distribution rebound effect after TPE due to the drug redistributing back into the blood after it has been removed [6]. This post-distribution rebound effect has been documented in drugs with high V_d, such as vancomycin, [11] phenytoin, [12, 13] and carbamazepine [14] as well as those with moderate to low V_d such as gentamicin [15] and tobramycin [16]. The results in the literature have reported that TPE has the most influence on drugs with a low V_d, regardless of the extent of protein binding [10]. Therefore, in light of the scarcity of evidence, volume of distribution should be one of the first parameters considered when determining if a drug will be removed with TPE.

Many drugs bind to proteins found in plasma, including albumin and alpha acid glycoprotein. Drugs that have high protein binding affinity, defined as > 80% of fraction bound (f_b), remain in the plasma and are therefore more

Table 1 Summary of the effect of therapeutic plasma exchange (TPE) on antimicrobials

Drug	fb (%) Red >80	Vd (L/kg) Red <0.2-0.3	Half-life (h) Red > 2h	T/OD	n	Time from last dose to TPE (h)	ΔC from TPE (post-TPE vs pre-TPE) %	TPE fraction elimination of body stores (%)	Drug Removed	Recommendation Comments Likely to be removed by TPE?
Acyclovir[29] (PK study)	9-32	0.69-0.8	2.5	T	3	1-3	56.6	2.5	No	No dose adjustment required
Amphotericin B deoxycholate[30, 31] (CR x 2)	90	4	α: 15-48h* β: 15 days	OD	2	36-72	5-63	NR	Possible	Higher concertation decline when TPE instituted closer to administration; High Vd suggest less elimination with chronic dosing; Administer after TPE
Amphotericin B liposomal[32] (CR)	95	0.1-0.16	7-153	T	1	38	40	NR	Yes	Supplementation post-TPE may be required as concentration fell below minimum inhibitory concentration
Ampicillin[33] (Ph II)	15-20	0.2-0.3	1-2	T**	15	2-10	35	NR	Possible	PK suggest no significant removal; possibly removed if given during TPE run
Cefepime[34] (Ph II)	20	0.26	2	T	9	1.5	NR	2.1-6.7	No	No dose adjustment required
Ceftazidime[35] (Ph II)	10-20	0.23	1-2	T	11	0.25-2	NR	4.6 - 6.9	No	No dose adjustment required; If given before TPE, allow time for distribution (~1-2h) before initiating TPE
Ceftriaxone[36, 37] (Ph II x 2)	83-96	0.1-0.2	5-9	T	22	0-15	26-53	5.7-25	Possible	Less removal if TPE after distribution phase (3h)
Chloramphenicol[38] (CR)	40	0.6-1	10-24**	OD**	1	29	See comment	NR	Possible	Concentration declined from 135 to <10 μg/ml with 8-10 exchanges; Half-life shortened from 80 (in-between runs) to 6.5h (during TPE) The study suggests possible removal
Dapsone[39] (CR)	70-80	1.5	28	OD	1	72	NR	2	No	Report suggests no dose adjustment required; give after TPE
Darunavir[40] (CR)	95	88 L	15	T	1	NR	60	1.5	Possible	Mean amount removed 9.3 mg of 600 mg dose; dose after TPE if possible
Ganciclovir[41] (CR)	1-2	0.5-0.8	3.5	T	1	NR	NR	8	No	No dose adjustment required
Gentamicin[15, 33, 42, 43] (Ph II x 4)	30	0.2-0.35	2	T**	41	2-12	26-62	5	Possible	Higher % of removal if TPE initiated right after the dose given; Administer dose after TPE if possible; TDM is available (For TDM, follow institutional protocols; EID: monitor 8h post or trough level; CD, monitor pre/post levels)
Pip-Tazo [44] (CR)	26-33	0.243	0.7-1.2	T	1	CI	NR	NR	No	TPE contributed to 7.1-10.9% of total clearance of the drug; no dosage adjustment necessary; for intermittent dosing, give dose after TPE
Tobramycin[16, 45, 46] (CR x 2, CS)	< 30	0.2-0.3	2-3	T	4	0.7–18	13.8-52.4	4 -11	Possible	Higher % of removal if TPE initiated right after the dose; Administer dose after TPE if possible; TDM is available (For TDM, follow institutional protocols; EID: monitor 8h post/trough level; CD, monitor pre/post levels)
Vancomycin[11, 47-50] (CRx4, Ph II)	55-70	0.4-1	4-6	T	16	5-134	8-55	4.5-8.1	Possible	Higher % of removal if TPE initiated right after the dose given; Administer dose after TPE if possible; TDM is available (For TDM, follow institutional protocols; monitor trough level or AUC)
Voriconazole[51, 52] (CR x 2)	58	4.6	6	T	2	0-2.5	0	2	No	No dose adjustment required
Zidovudine[53](CR)	25-38	1-2.2	0.5-3	T	1	4	50-71	≤ 1	No	No dose adjustment required

PK parameters may change in critical illness, and they were added here to show the possibility of removal by TPE

AUC area under the concentration–time curve, *ΔC* concentration change, *CD* conventional dosing, *CI* continuous infusion, *CR* case report, *CS* case series, *EID* extended interval dosing, *fb* fraction bound, *n* number of subjects, *NR* not reported, *OD* report of overdose, *Ph II* phase II study, *Pip-Tazo* piperacillin tazobactam, *PK* pharmacokinetic, *T* report of therapeutic dosing, *TDM* therapeutic drug monitoring, *Vd* volume of distribution

**α* is distribution half-life while β is elimination half-life; **data from neonates

susceptible to removal by TPE than drugs that are considered to have low or negligible protein binding [17]. It is important to keep in mind that the extent of drug protein binding is not uniform among patients and that reported values given in drug monographs are based on a population average of healthy subjects or non-critically ill patients. Many factors can affect drug protein binding which consequently can change the effect of TPE in regard to drug removal and efficacy [5]. To illustrate, disease states, such as renal and hepatic failure, can cause decreased levels of albumin which, in turn, will cause decreased binding of acidic drugs [18]. Factors such as trauma, surgery and infections resulting in inflammation may cause an increase in the level of alpha acid glycoprotein and cause increased binding of basic drugs [19]. Additionally, factors affecting protein metabolism, such as nutritional status, can also have an effect on plasma protein concentrations [20]. Furthermore, the administration of FFP or albumin as replacement fluids after TPE can affect drug distribution and pharmacodynamics [2]. While drugs bound to plasma proteins during TPE will be removed during the process, TPE without protein replacement can increase the free fraction of the drug after TPE due to redistribution. The free drug fraction is the pharmacologically active form, and factors that alter its concentration have the potential to change the drug's efficacy and adverse reactions. On the other hand, albumin replacement results in more bound drug and potential reduction in drug efficacy [21].

Another factor to consider is the multi-compartmental distribution of drugs. Drugs that exhibit multi-compartmental pharmacokinetics display a distribution phase and elimination phase as depicted in Fig. 1. The initial decline in the distribution phase represents the drug leaving the plasma (central) compartment, either by entering into a tissue (peripheral) compartment or through endogenous elimination, until a drug plasma equilibrium occurs with peripheral tissues. The elimination phase occurs after this equilibrium is reached, and the drug must diffuse back from the peripheral compartment to the plasma in order to be eliminated. For drugs that exhibit multi-compartmental kinetics, it is important to recognize that the timing of TPE will have an effect on drug removal. If TPE is implemented during the distribution phase of the drug, the drug will be more susceptible to removal than if TPE is run during the elimination phase, as a much more considerable drug concentration will be present in the plasma [5, 7, 22].

Table 2 Summary of the effect of therapeutic plasma exchange (TPE) on antiepileptic drugs

Drug	fb (%) Red >80	Vd (L/kg) Red <0.2	Half-life (h) Red > 2h	T/OD	n	Time from last dose to TPE (h)	ΔC from TPE (post-TPE vs pre-TPE) %	TPE fraction elimination of body stores (%)	Drug Removed	Recommendation Comments Likely to be removed by TPE?
Carbamazepine[14, 54-58] (CR x 6)	70-80	0.59-2	17-40	T (1); OD (5)	6	12-32	4-46.8	5.7	Possible	In overdose cases, small amount is removed as Vd is large. Transient reduction of concentrations post TPE may be of clinical relevance. Dose adjustment not likely required; TDM available.
Clobazam[59] (CR x 1)	80-90	100 L	10-30	T	1	0-4	25-29	NR	Possible	Transient reduction of concentrations post TPE may be of clinical relevance. Estimated 3-4mg removed/TPE session
Levetiracetam[59, 60] (CR x 2)	< 10	0.5-0.7	6-8	T	2	4-12	17-28	5	Possible	Negligible amount removed. Dose adjustment not likely required. Transient reduction of levels post TPE may be of clinical relevance. Administer dose after TPE if possible
Oxcarbazepine[61] (CR)	40	0.75	9	T	1	1	NR	5.8	Possible	No dose adjustment required; Transient reduction of concentrations post TPE may be of clinical relevance.
Phenobarbital[62, 63] (CR x 2)	75-95	0.6	7 d	T	2	118	NR	3-38	Possible	No dose adjustment required; Allow time for distribution before TPE
Phenytoin[12, 13, 64-70] (CR x 7 CS x 2)	88-92	0.52-0.78	7-42	T (8); OD (1)	15	1.5->72	Total: 19-43 % 1h post: 15-19.5% (rebound) Free: 0-49.5 % 1 h post: 3.4-18%	3-10	Possible	Negligible amount removed. Dose adjustment not likely required. May see larger changes in total levels immediately post-TPE, but free levels do not change much. Also, redistribution and level rebound seen; TDM available (Monitor trough levels; post TPE levels within 1h may be needed if poor seizure control while on TPE; monitor free levels if available)
Topiramate[59] (CR)	13-17	0.55-0.8	21	T	1	4-8	21-26	NR	Possible	Transient reduction of levels post TPE may be of clinical relevance. Administer dose after TPE if possible; Estimated 47 mg removed per one TPE session
Valproic acid[63, 71] (CR x 2)	80-90	0.3-0.7	9-19	T	2	3.5	45	7-8.6	Possible	Likely no dosage adjustment required, but should check trough levels and adjust as required unless poor control around TPE session (post TPE levels within 1h may be needed); monitor free levels if available.

Pharmacokinetic parameters may change in critical illness and added here to show the possibility of removal by TPE

CR case report, *CS* case series, *ΔC* concentration change, *fb* fraction bound, *n* number of subjects, *NR* not reported, *OD* report of overdose, *T* report of therapeutic dosing, *TDM* therapeutic drug monitoring, *Vd* volume of distribution

Table 3 Summary of the effect of therapeutic plasma exchange (TPE) on antithrombotics

Drug	fb (%) Red >80	Vd (L/kg) Red <0.2	Half-life (h) Red > 2h	T/OD	n	Time from last dose to TPE (h)	ΔC from TPE (post-TPE vs pre-TPE) %	TPE fraction elimination of body stores (%)	Drug Removed	Recommendation Comments Likely to be removed by TPE?
Aspirin[72] (PK)	80-90	0.1-0.2	3-6	T	6	1.5	25	7-32	Possible	Anti-inflammatory doses of aspirin were given; possibly removed; monitor clinically. For antiplatelet dosing, give after or 2h before TPE.
Apixaban[73] (CR)	87	21 L	8-15	T	1	29	58-71 (Anti-Xa)	NR	Yes	Removed by TPE; dose after run
Dalteparin[74, 75] (CR, Ph II)	N/A	0.04-0.06	3-5	T	12	0-4.5	40 (dalteparin) 53-75 (Anti-Xa)	NR	Yes	TPE removes free drug, and drug bound to antithrombin. Give dose after TPE. TPE also removes coagulation factors.
Enoxaparin[60, 76] (CR x 2)	N/A	4.3 L	4.5-7	T	2	7-23	100 (Anti-Xa)	NR	Yes	Give dose after TPE. TPE also removes coagulation factors.
Heparin[77] (CS)	Binds to ATIII	0.07	0.7-2.5	T	4	CI	38-47 (Anti-Xa)	NR	Yes	Study maintained stable anti-Xa by increasing heparin rate by 65% during TPE. Affected by FFP vs albumin replacement. TPE also removes coagulation factors. Heparin may be ineffective if excessive ATIII is removed.
Rivaroxaban[78] (CR)	92-95	50 L	5-9	T	1	12	47.5 (Anti-Xa)	NR	Yes	TPE also removes coagulation factors; administer after TPE
Warfarin[79, 80] (CR, Ph II)	99	0.14	20-60	T (1); OD (1)	9	23	CR: INR 48 % (2.37 → 1.23) Ph II: See note	NR	Possible	CR: Received FFP as replacement, which can lower INR. Vitamin K also given. From PK characteristics, TPE should remove warfarin Ph II: Factor II activity mean decrease of 1% (decrease 11% - increase 16%). Fibrinogen median decrease of 71 mg/dl (3-127 mg/dl) INR mean increase of 0.02 (decrease 0.16 – increase 0.02) Monitor INR

PK parameters may change in critical illness and added here to show the possibility of removal by TPE

AT III antithrombin III, *CR* case report, *CS* case series, *ΔC* concentration change, *fb* fraction bound, *FFP* fresh frozen plasma, *INR* international normalised, *n* number of subjects, *NR* not reported, *OD* report of overdose, *Ph II* phase II study, *PK* pharmacokinetic, *T* report of therapeutic dosing, *Vd* volume of distribution

Table 4 Summary of the effect of therapeutic plasma exchange (TPE) on cardiovascular drugs

Drug	fb (%) Red >80	Vd (L/kg) Red <0.2-0.3	Half-life (h) Red > 2h	T/OD	n	Time from last dose to TPE (h)	ΔC from TPE (post-TPE vs pre-TPE) %	TPE fraction elimination of body stores (%)	Drug Removed	Recommendation Comments Likely to be removed by TPE?
Amiodarone[81, 82] (CR x 2)	>96	66	26-107 d	T	2	18-31 d	Negligible	NR	No	Not likely to be removed owing the very high volume of distribution; no dosage adjustment required
Amlodipine[83, 84] (CR x 2)	93	21	30-50	OD	2	36	48-73	NR	Possible	Potential option in OD. Would expect redistribution as high Vd, therefore may not need to re-dose post TPE. Monitor clinical status of patient.
Carvedilol[85] (CR)	>98	115 L	7-10	OD	1	NR	50	NR	Possible	Potential option in OD. Would expect redistribution as high Vd, therefore may not need to re-dose post TPE. Monitor clinical status of patient.
Digoxin[79, 86-91] (CR x 3, CS x 4)	25	5-8	36-48	OD	17	0-72	14-75	0.1-3.4	No	Not significantly removed. Potentially significant amount of digoxin removed if given DigiFab given prior to TPE
Diltiazem[92] (CR)	70-94	3-13	3-4	OD	1	5-16	92	NR	Possible	Possibly removed. Monitor response of the patient and determine if supplemental dose needed.
Disopyramide[93] (CR)	50-65	0.8-2	4-10	T	1	2.5	Initial 60% rebound to 7% 1.5h post TPE	2.7	No	Not significantly removed. No dose adjustment needed
Mexiletine[94] (CR)	60	5-7	10-12	T	1	CI	NR	See note	No	4.5 mg removed from 60 mg/h infusion; less likely to be removed
Propafenone[79] (CR)	95	1.1	2-32	OD	1	23	NR	NR	Possible	Symptomatic improvement; not clear if related to TPE
Propranolol[95-97] (CR x 3)	90	4	IR: 3-6 ER: 8-10	T (1); OD (2)	3	3-36	74-80	NR	Possible	Potential option in OD. Would expect redistribution as high Vd, therefore may not need to re-dose post TPE. Monitor clinical status of patient.
Verapamil[96, 98, 99] (CR x 3)	90	3.89	IR: 3-7 ER: 12	OD	3	2.5-36	58-69	NR	Possible	Potential option in OD. Would expect redistribution as high Vd, therefore may not need to re-dose post TPE. Monitor clinical status of patient.

PK parameters may change in critical illness, and they were added here to show the possibility of removal by TPE

ΔC concentration change, *CI* continuous infusion, *CR* case report, *CS* case series, *ER* extended release formulation, *fb* fraction bound, *IR* immediate release formulation, *n* number of subjects, *NR* not reported, *OD* report of overdose, *T* report of therapeutic dosing, *Vd* volume of distribution

Table 5 Summary of the effect of therapeutic plasma exchange (TPE) on immunosuppressants and chemotherapy

Drug	fb (%) Red >80	Vd (L/kg) Red <0.2-0.3	Half-life (h) Red > 2h	T/OD	n	Time from last dose to TPE (h)	ΔC from TPE (post-TPE vs pre-TPE) %	TPE fraction elimination of body stores (%)	Drug Removed	Recommendation Comments Likely to be removed by TPE?
Basiliximab[100, 101] (CR x 2)	N/A	4.5-12.7 L	4-10.4 days	T	2	6	65-72	46	Yes	May need to give supplemental dose to concentrations above IL-2R saturation levels (0.2 ug/mL)
Cisplatin[22, 102-105] (CR x 5)	90	11-12 L/m^2	Platinum: ≥5days	OD	5	5h-22 days	24-93	NR	Yes	Likely removed, would expect redistribution as high Vd, therefore may not need to re-dose post TPE. Monitor clinical status of patient.
Cyclosporine[106-110] (CR x 5)	90-98	4-6	8.4	T (2) OD (3)	5	2-13	6-36	0.2-0.3	No	Drug is distributed to erythrocytes; in a case report of OD, whole blood exchange and TPE reduced concentration by 95%
Methotrexate[111-113] (CR x 3)	50-60	0.4-0.8	3-15	OD	3	24-120	NR	0.9-2	No	TPE may promote beneficial effect in renal function, however, TPE largely ineffective in removing drug
Mycophenolate[114, 115] (CR, CS)	97	MMF: 3.6 MPA: 54 L	MMF: 18 MPA: 8-16	T	3	4-7.5	19-70	0.44-2.8	No	Higher % of removal if TPE initiated right after the dose given or within distribution phase (<4 hrs prior to TPE); Administer dose after TPE if possible
Natalizumab[116] (Ph II)	N/A	3.8-7.6 L	7-15 days	T	12	10-14 days	65-82	NR	Yes	TPE may help with reducing levels if patient develops PML. α4-integrin saturation: May need natalizumab below certain threshold (i.e. 1 ug/mL) to achieve reduction in receptor saturation; Uncertain if drug removal affects clinical response
Prednisone[117] (CS)	90-95	0.6-0.7	3-4	T	2	0.42-1.16	NR	0.83	Possible	Not likely removed; dose post TPE
Rituximab[118-121] (CR x 2, Ph II x 2)	N/A	3.1-4.5 L	18-32 days	T	53	24-165	45-77	9-54	Yes	Drug is likely removed, administer after TPE if possible. Uncertain if drug removal affects clinical response
Tacrolimus[122-124] (CR, CSx2)	99	0.85-1.41	23-46	T	6	1-7.5	NR	NR	No	Drug is distributed to erythrocytes and negligibly removed by TPE. Monitor trough levels.
Vincristine[125, 126] (CR, CS)	50-80	215 L/1.73^2	19-155	OD	4	6	8-72	NR	Possible	Potential option in OD. Would expect redistribution as high Vd

PK parameters may change in critical illness, and they were added here to show the possibility of removal by TPE

ΔC concentration change, *CR* case report, *CS* case series, *ER* extended release formulation, *fb* fraction bound, *IR* immediate release formulation, *MMF* mycophenolate mofetil, *MPA* mycophenolic acid, *n* number of subjects, *NR* not reported, *OD* report of overdose, *Ph II* phase II study, *PML* progressive multifocal leukoencephalopathy, *T* report of therapeutic dosing, *Vd* volume of distribution

Table 6 Summary of the effect of therapeutic plasma exchange (TPE) on other drugs

Drug	fb (%) Red >80	Vd (L/kg) Red <0.2-0.3	Half-life (h) Red > 2h	T/OD	n	Time from last dose to TPE (h)	ΔC from TPE (post-TPE vs pre-TPE) %	TPE fraction elimination of body stores (%)	Drug Removed	Recommendation Comments Likely to be removed by TPE?
Acetaminophen[127] (Ph II)	10-25	1	2	T	5	1	NR	4	No	Not likely removed; no dose adjustment needed
Amitriptyline[128-132] (CR x 4, CS)	95	19	13-36	OD	7	2-39	60-92	NR	Possible	Potential option in OD
Diclofenac[127] (Ph II)	>99	1.3-1.4	2	T	4	1	NR	17	Possible	Possible; dose post TPE
Ibuprofen[133] (CR)	99	0.11-0.19	2	OD	1	9	35	NR	Possible	Possibly removed
Quinine[134] (CR)	85-95	1.8	10-13	OD	1	45	NR	NR	Possible	Percent fraction removed is not clear and patient was on concomitant peritoneal dialysis; 8.5 mg was recovered in waste plasma. Give after TPE if possible.
Theophylline[135, 136]	40	0.45	6-12	T* (1);	4	3->15	16.9-36	NR	Possible	Give after TPE if possible

Pharmacokinetic parameters may change in critical illness, and they were added here to show the possibility of removal by TPE

ΔC concentration change, *CR* case report, *CS* case series, *fb* fraction bound, *n* number of subjects, *NR* not reported, *OD* report of overdose, *Ph II* phase II study, *T* report of therapeutic dosing, *Vd* volume of distribution

*Data from neonates

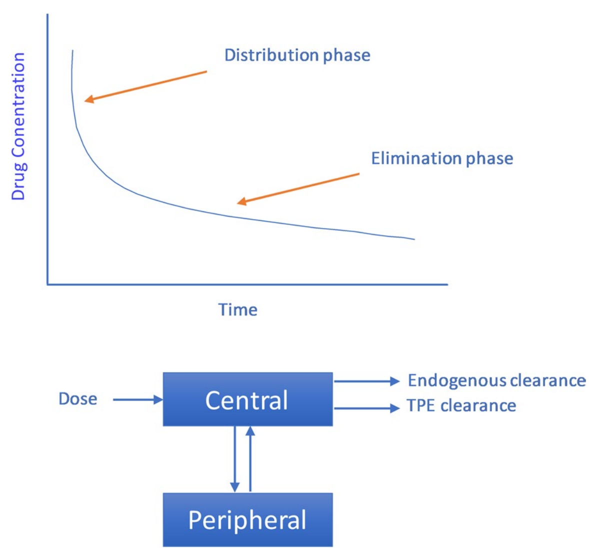

Fig. 1 Drugs that exhibit multi-compartmental pharmacokinetics display a distribution phase and elimination phase. The initial decline in drug concentration (the distribution phase) represents the drug leaving the plasma (central) compartment, either by entering into a tissue (peripheral) compartment or through endogenous elimination, until a drug plasma equilibrium occurs with peripheral tissues. The elimination phase occurs after this equilibrium is reached, and the drug must diffuse back from the peripheral compartment to the plasma in order to be eliminated. If therapeutic plasma exchange is implemented during the distribution phase of the drug, the drug will be more susceptible to removal than if TPE is run during the elimination phase, as a much more considerable drug concentration will be present in the plasma

Metabolism and Excretion: Clearance and Half-Life

The clearance of a drug can be defined as the volume of blood cleared from the drug per unit time. When drugs have short half-lives [21] or are rapidly metabolized [23], they will be cleared from the body quickly and will therefore be less susceptible to removal by TPE. It has been suggested that if endogenous clearance is < 4 mL/min [24] or drug metabolism is impaired, due to factors such as kidney or liver dysfunction, there is a greater likelihood that the drug will be removed via TPE as more drug would be present in the system. While clearance is an important parameter to take into consideration, it is difficult to measure in practice and not readily available for clinicians in drug product monographs. Therefore, it is much easier to use the half-life of the drug as it is often available from the drug monograph, as a surrogate marker of clearance as half-life is dependent on both clearance and V_d. As the approximate duration of most TPE procedures is 2 h, drugs that have half-lives longer than 2 h are more susceptible to TPE removal [21].

TPE-Specific Factors

Related to the pharmacokinetic distribution of the drug, the time between the drug dose and TPE initiation is a major factor in the amount of drug extracted with TPE [7]. Evidence has suggested that if given enough time to fully distribute, drug elimination via TPE will be significantly decreased due to much lower concentrations of the drug in the plasma regardless of V_d or protein binding [7]. With this in mind, if TPE is not being used for the purpose of drug extraction (such as in the case of a drug overdose), whenever possible, it is beneficial to dose the drug after TPE has completed. If dosing after TPE is not an option, the drugs distribution half-life, if available, can be used to help estimate the amount of drug distributed to the extravascular space at the time of TPE initiation [25]. It is worth noting that a distribution half-life and

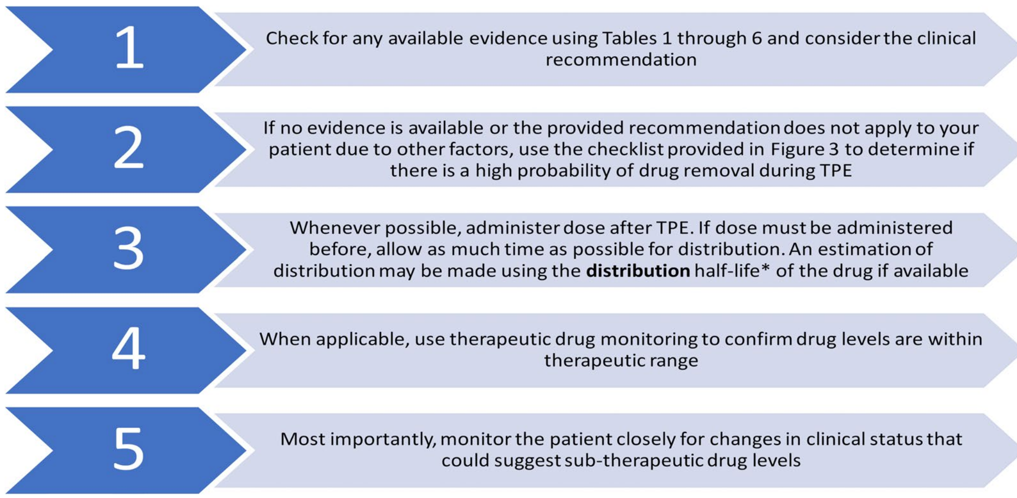

Fig. 2 Stepwise approach on managing drugs in patients undergoing therapeutic plasma exchange (TPE). *, the distribution half-life is shorter than the elimination half-life. If the distribution half-life is not available, we suggest using the elimination half-life as a surrogate to gauge drug distribution. For example, for a drug with an elimination half-life of 3 h and administered by an intravenous bolus, we suggest waiting for at least 3 h following dose administration before TPE initiation. Waiting for 3 h (one half-life) will allow at least 50% of the drug to be distributed to the tissues

elimination half-life of a drug are not representative of the same mechanism. However, distribution half-life values are not readily available for clinicians in drug product monographs and are mainly obtained from pharmacokinetic reports. Since distribution half-lives are always shorter than the elimination half-lives, we suggest using the elimination half-lives as surrogate to gauge drug distribution if distribution half-lives are not available.

Other TPE-specific factors relating to the amount of drug removed are the duration, volume of exchange and frequency of the procedure. The procedure typically occurs over 2–3 h [1] while the frequency is largely dependent on the underlying reason for the treatment. Exchanges of 1.0, 1.5, and 2.0 times plasma volume (~60 ml/kg) correlate with a removal of approximately 63%, 78% and 86% of plasma contents, respectively [2]. An increase in duration, frequency or volume of exchange all increases the likelihood of a greater amount of drug extracted.

Clinical Recommendations

Given that the majority of the studies published on this topic are based of case reports, which correspond to a very low level of evidence [26], no definitive guidelines have been created regarding drug removal while undergoing TPE. Additionally, drug extraction by TPE should not be categorized into absolute “yes” or “no” terms as the clinical status of the patient in question may be different than those previously reported [6]. As such, we suggest Figs. 2 and 3 to provide a step-by-step guide for clinicians to apply practical knowledge and form applicable clinical judgments surrounding the topic. The clinical status of the patient needs to be kept at the forefront when deciding if a drug requires a dose adjustment due to TPE. If a transient change in drug concentration is of clinical relevance, such as antiepileptic drugs or concentration dependant antibiotics, then more significance should be given to the potential removal of the drug compared to those in which a transient change has less of an effect on clinical outcome.

Therapeutic drug monitoring (TDM), if available, is a very useful tool to use if TPE is suspected to influence the clinical response to the drug. TDM protocols for drugs with narrow therapeutic range in patients undergoing TPE are similar to those not on TPE. For example, if monitoring trough levels is recommended such as in case of antiepileptic drugs, clinicians still need to monitor trough levels in patients undergoing TPE. However, additional levels may be required in patient-specific scenarios. For example, although phenytoin and valproic acid are less likely to be significantly removed by TPE, patients may experience poor seizure control due to transient reduction in drug concentrations post-TPE. A post-TPE level (within 1 h) may be required to determine if an extra dose is needed to be given before subsequent TPE runs to maintain seizure control. For drugs that are highly protein bound, it is recommended to monitor free levels, if available.

Limitations

A review of the current literature mostly consists of case reports or case series investigating the effects of TPE on

	Yes	No	Unsure
Is the volume of distribution (Vd) < 0.2 L/kg?			
Is protein binding (fb) > 80%?			
Is the half-life (t½) > 2h?			
Will TPE start during the distribution phase of the drug OR will the drug be dosed immediately prior to or during TPE ?			
Does the patient have dysfunction of a drug elimination organ?			
Are transient changes in concentration of clinical relevance?			
Is there any new evidence that suggests removal of the drug by TPE?			

Mostly "Yes"
- It is likely that some of this drug will be removed with TPE: administer dose after TPE when available; dose supplementation after TPE may be required; use TDM when possible

Mostly "No"
- It is unlikely that this drug will be removed using TPE: no drug adjustment is required

Unsure
- Use TDM when possible; closely monitor patient for changes in clinical status that could suggest sub-therapeutic drug levels

Fig. 3 Checklist to determine how likely drugs are removed by therapeutic plasma exchange (TPE)

drug levels mostly related to accidental or intentional drug overdose [27]. Many of these case reports are often studied in patients undergoing TPE for different diseases and in conjunction with other treatments as well [28]. In addition, many case reports utilized TPE with different durations, volumes, frequency and initiation time of TPE along with different measured endpoints resulting with the incomparable results [10]. Furthermore, drugs behave differently under different clinical contexts, and therefore, caution is recommended when applying previous findings to current clinical scenarios if the clinical context is markedly different. In addition, reported V_d, protein binding and half-lives are based on population averages in non-critically ill adult populations. These parameters can be notably different in critically ill patients, which are often those who require TPE. These limitations make it difficult to draw definitive conclusions regarding the amount of drug removed via TPE.

Conclusion and Future Directions

TPE is a procedure that can influence the disposition of drugs in circulation. The current review suggests that when predicting drug removal via TPE, patient-specific parameters and clinical status should be taken into consideration before making a clinical decision. Whenever possible, administer the medication after TPE. Additionally, the use of V_d, extent of protein binding and the half-life of the drug combined with available evidence can further assist in determining the likelihood of drug removal by TPE. Future studies conducted in a standardized manner with detailed information are needed to further enhance our knowledge of the effects of TPE.

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Acknowledgements

Janice Kung, Librarian at John W. Scott Library, for assisting with the keywords' selection and search strategy.

Author contributions

SHM and TH designed the study. MEDLINE and EMBASE search and data extraction were conducted by EC and confirmed by JB and SHM. Additionally, TH and SC conducted MEDLINE and PubMed search and data extraction, independent from the other search. All authors wrote and revised the manuscript.

Source of Support

There is no funding associated with this work.

Conflict of interest

The authors declare that they have no conflicts of interest

Ethics approval and Informed Consent

This article does not contain any studies with human participants or animals performed by any of the authors. For this type of work, formal consent is not required.

Publisher's Note

Published online: 22 May 2020

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